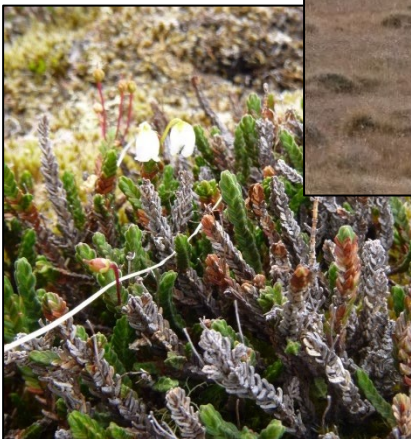




Nunavik Nickel Project

Annual report Biodiversity conservation - 2023



INTRODUCTION

Canadian Royalties Inc (CRI) mines nickel and copper ores at its Nunavik Nickel Project (NNiP) mining complex. This mining project is located approximately 80 km west of Kangiqsujuaq and about 140 km southeast of Salluit in Northern Quebec. PNNi began in 2008 with the discovery of the Mesamax, Expo, Méquillon and Ivakkak deposits. The Allammaq and Puimajuq deposits were subsequently added to the global certificate of authorization no. 3215-14-007 issued by the Quebec government.

As a member of the Quebec Mining Association, CRI adheres to the *Mining* Association of Canada's *Towards Sustainable Mining* (TSM) initiative. CRI implements the biodiversity conservation management protocol by integrating it into its activities and into the PNNi Environmental Management Program. CRI is also committed to ensuring that its actions are consistent with the protocol's strategic framework, presented in the appendix to this report.

An Environmental and Social Impact Assessment (ESIA) was carried out in 2007 to characterize the receiving environment, including the biological environment. Inventories of vegetation, aquatic fauna, herpetofauna, avian fauna and mammals were carried out, and their potential impacts assessed. An environmental monitoring program (EMP) and a set of mitigation measures were developed and have been implemented for several years.

CRI has also drawn up a Plant and Wildlife Protection Plan (PPFF). CRI's objective is to carry out the protection and monitoring measures set out in the PPFF on an annual basis, and the present document reports on these measures for 2023.



Stéphane Twigg, Environmental Manager

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1. COMPLETED ACTIONS

Theme 1: Improving knowledge assessment

Action 1: Cycle 4 environmental effects monitoring (EEM) study and monitoring of metal concentrations in fish flesh from the Puvirnituk River

The environmental effects monitoring (EEM) study for Expo mine effluent consisted of a fish community survey, complemented by a benthos characterization, and was carried out in July 2023 by the firm Aecom. Fish were caught in an area exposed to effluent one (a tributary of the Puvirnituk River) and in a reference area, in order to highlight any differences in population growth, reproduction, survival or condition, and thus determine whether mining effluent had an effect on fish. The sentinel species selected was Arctic char (*Salvelinus Alpinus*). To this end, morphometric measurements (length, body weight, age) and flesh samples for metal analysis were taken and statistically processed. Characterization of the benthos on which the fish feed provides additional information to the observations made on the fish themselves.

The EEM study plan has been approved by *Environment and Climate Change Canada* (ECCC), and was carried out in accordance with the requirements of ECCC's *Metal and Diamond Mining Regulations*. The interpretive report is currently being written and will be completed in February 2024. This study is also included in PNNi's EMP (follow-ups 8, 9 & 16); the results are forwarded to MELCCFP¹ and the Inuit communities in the annual EMP report.



This is the 4th ESEE study carried out by CRI and the knowledge acquired during this study are added to those accumulated in 2014, 2017 and 2020.

Status : Completed

Study details : AECOM. 2022. Cycle 4 Environmental Effects Monitoring (EEM) Plan - Biological Monitoring Study for Canadian Royalties Inc. Nunavik Nickel Project. Report submitted to Canadian Royalties Inc. and Environment and Climate Change Canada. 50 pages and appendices.

Action 2: Monitoring the population and mercury concentration in the flesh of fish in Lac du Bombardier

The population of Bombardier Lake was monitored in July 2023 by the firm Aecom; the purpose of this monitoring was to verify whether the raising of the water level and the presence of the water intake had any effect on the recruitment of the fish species present. It is complemented by a follow-up of the mercury concentration in the flesh of the lake's fish, also to determine whether the raising of the water level has led to an increase in this concentration. To determine the effects, if any, of raising water levels, fishing and diving on the

¹ MELCCFP: Ministère de l'Environnement, de la Lutte aux Changements Climatiques, de la Faune et des Parc.

are carried out. Fishing stations and spawning grounds are characterized (granulometry, physico-chemistry, etc.) and fishing effort calculated. Morphometric measurements are taken on the fish used for mercury analyses.

This follow-up is carried out at intervals specified in the PES (follow-up 10 & 11); the results are forwarded to the MELCCFP and the Inuit communities in the annual PES report.

Status : Completed

Action 3: Continuation of the register of caribou sightings in the Méquillon sector and inclusion of worker sightings

From June to the end of August, the Department of the Environment continues to keep a caribou observation register, covering the migration period. The number of individuals, their description (male, female, calf) and direction of movement are recorded. In 2023, workers in other departments have also been encouraged to record their sightings, in order to build up a more complete picture of their present situation.

This follow-up is included in the PNNi PSE (follow-up 18); the results are transmitted to the MELCCFP and the Inuit communities in the annual PSE report.

Status : Completed



Theme 2: Preventing and protecting biodiversity

Action 4: Monitoring the free movement of fish in the culverts of PNNi



CRI's environmental technicians monitored the free movement of fish through the culverts at PNNi on three occasions, between June and September. To do this, various measurements were taken upstream and downstream of the culvert, including water height and velocity, and overall observations were made to identify potential obstacles to fish passage (e.g. rocks that had fallen from the road). Water level and velocity measurements are compared with values from

the culvert from downstream to upstream and thus circulate freely between the water bodies surrounding the roads in PNNi.

This follow-up is included in the PNNi PSE (follow-up 12); the results are transmitted to the MELCCFP and the Inuit communities in the annual PSE report.

Status : Completed

Action 5: Monitoring of Arctic char spawning grounds

Monitoring of the Arctic char spawning ground located near

the

of the Puimajuq road crossing was carried out by Aecom in the summer of 2023. To this end, stations (10 quadrats measuring 1 m x 1 m) were characterized to assess substrate, current speed and depth, thus confirming that the site remains suitable for fish spawning despite the development of the crossing. CRI's environmental technicians also take weekly samples to measure the concentration of suspended solids in the water.

This follow-up is included in the PNNi PSE (follow-up 14); the results are transmitted to the MELCCFP and the Inuit communities in the annual PSE report.

Status : Completed



Action 6: Monitoring of loggerhead damselfish colonies



Three species of special-status plants were observed during the surveys in the Allammaq sector, belonging to the dragonfly genus (*Draba subcapitata*, *D. micropetala* and *D. corymbosa*). Colony status was monitored by Aecom in summer 2023. To this end, in each colony, draba plants were counted and any abnormal condition of plant structures was noted (e.g. yellowing of leaves, absence of flowers or fruit, etc.).

This follow-up is included in the PNNi PSE (follow-up

20); the results are reported to the MELCCFP and the Inuit communities in the annual PSE report.

Status : Completed

Action 7: Drawing up a wildlife protection plan and updating the Wildlife Management procedure

In 2023, CRI drew up a Plant and Wildlife Protection Plan (PPFF), with the aim of making a positive contribution to biodiversity conservation. The PPFF integrates in a single document the protection, monitoring and follow-up measures that need to be applied to avoid and reduce the negative effects of PNNi on flora and fauna, thus facilitating their dissemination and application. The components of interest identified are :

- | | |
|--|---|
| ▪ Threatened or vulnerable plant species ; | ▪ The gray wolf, the black bear and the polar bear; |
| ▪ Wetlands and hydric environments ; | ▪ Caribou ; |
| ▪ Aquatic fauna ; | ▪ Arctic fox and red fox; |
| ▪ Avian fauna and nest protection; | ▪ Collisions with wildlife. |

The Wildlife Management procedure has been revised to better reflect the objectives of the PPFF, and a decision chart detailing the rules of conduct for road vehicles in the presence of caribou near PNNi roads has been integrated.

Status : Completed

Action 8: Installation of visual markers around plants with precarious status

In the summer of 2023, CRI installed visual cues around plants of precarious status that had been inventoried in recent years. More specifically, these are specimens of *Draba micropetela* (TR4 crossing sector) and sulphur buttercup (*Renonculus sulphureus*; traverse TR6). The purpose of the visual markers is to prevent encroachment on the plants (e.g. by snowmobiles in winter for mining exploration activities, or during the construction of a new mine).



construction work. The installation of these markers is provided for in the PNNi Fauna and Flora Protection Plan, and will therefore be renewed in the event of any new sightings of plants of precarious status.

Status : Completed

Action 9: Acquisition of waste bins less attractive to wildlife



With a view to improving wildlife management, CRI acquired new outdoor waste bins and also modified existing ones to make food waste less accessible to wildlife. The aim was to limit the attractiveness of the mining camp to wildlife, and thereby reduce the risks posed to them by interaction with humans.

Action 10: Workshops and awareness-raising capsules for employees

In 2023, the environmental departments delivered twelve *toolbox* workshops to employees in the various departments and distributed four capsules by e-mail or poster, all on the theme of biodiversity protection. Topics covered included driving rules when caribou are present near roads, and the importance of not feeding wildlife.

General environmental training also continued in 2023. This training is mandatory for all CRI workers and contractors, and one of its components covers

PNNi's biodiversity, with the aim of raising awareness of its importance and fragility, despite its low profile in our northern context.

Topic 3: Impact mitigation and compensation measures

Action 11: Advancing the Inuit Community Environmental Improvement Program (ICEEP)

CRI compensates for wetland encroachment related to NNNi activities through an Inuit Community Environmental Enhancement Program (ICEEP), in partnership with the Kativik Regional Government (KRG). In 2023, the KRG restored three sandpits in Puvirnituq, identified by the village council and in collaboration with local Inuit workers. An innovative soil reshaping technique is also being applied to certain sections of the sandpits, to facilitate revegetation. Revegetation will be monitored over a three-year period, including an experimental component for the planting of bioroutes and fruit-bearing plants.

Thus, CRI financially supports these projects, which are by and for the communities. The KRG is continuing discussions with the villages of Kangiqsujuaq and Salluit regarding the continuation of these projects and the development of new ones.

This follow-up is included in the PNNi PSE (follow-up 19); the results are transmitted to the MELCCFP and the Inuit communities in the annual PSE report.

Status : In progress

Action 12: Renewal of the installation of bird-scaring devices on MCPs CRI has

installed bird-scaring devices in the vicinity of the mine water collection basins at the Expo, Mesamax and Méquillon sites.

Status : Completed



Action 13: Creation of a weir downstream of the TR3 crossing

In 2022, monitoring of fish passage through the culverts revealed that a gap between the invert at the downstream end of one of the TR3 crossing culverts and the bed of the watercourse could constitute an obstacle to fish passage. A weir was therefore built downstream of the crossing to slightly raise the water level and facilitate fish passage. The layout plans and work methods had been approved in advance by Fisheries and Oceans Canada and the MELCCFP.



Status : Completed

Action 14: Reconstruction of the riverbed and creation of a fish passage channel at crossing TR2

In 2023, CRI undertook the reconstruction of a stream bed and the creation of a channel in one of the culverts at the TR2 crossing, to facilitate fish passage. The project plans and work methods had been approved in advance by Fisheries and Oceans Canada.



Status : Completed

Theme 4: Stakeholder consultation, information and involvement

Action 15: Information program in Inuit communities

Two representatives from CRI's Environment Department met with the three Inuit villages that will be signing the Nunavik Nickel Agreement between January and May 2023 (Kangiqsujaq, Salluit and Puvirnituq), to present the Environment Department's activities and the results of the environmental monitoring program, and to gather any environmental concerns.

Status : Completed

This follow-up is included in the PNNi PSE (follow-up 35); the results are transmitted to the MELCCFP and the Inuit communities in the annual PSE report.

Action 16: Consultations on phase 2 mining projects in PNNi - wildlife component

In 2023, consultations continued with stakeholders through the committees provided for in the Nunavik Nickel Agreement (Impact and Benefit Agreement) concerning the development of new identified mining projects and the extension of certain existing projects. Stakeholders are invited to express their concerns at these meetings, particularly with regard to flora and fauna. The study of proposed mitigation measures, including those relating to flora and fauna, continued in 2023.

Status : In progress

Theme 5: Support for research and development

Action 17: Contribution to Caribou Ungava

With a view to continuously improving its practices, CRI has entered into a partnership with Caribou Ungava, a research program focusing on the ecology of migratory caribou populations. CRI is participating in a research project on the impact of linear structures, such as mining roads, on caribou behaviour, by providing financial contributions and logistical support for research work. A team of researchers was hosted by CRI and spent four weeks in the PNNi and Deception Bay area in the summer of 2023, and will do so again in 2024 and 2025.

Status : In progress

Action 18: Support for MELCCFP representatives to locate a peregrine falcon

CRI welcomed representatives of the MELCCFP to its facilities at the end of June 2023, with the aim of spotting a peregrine falcon that had been observed in the fall of 2022. Morning and evening helicopter transportation was provided. Assisted by an Inuit employee and another CRI employee, they were able to capture the falcon and fit it with a transmitter to continue satellite tracking of the bird.

Status : Completed

Action 19: Data sharing with Projet Biodiversité Québec

In 2023, CRI met with representatives of the Projet Biodiversité Québec, affiliated with the University of Sherbrooke. Data from floristic inventories carried out by CRI over the past few years have been transmitted to them, in order to contribute to the Quebec Biodiversity Atlas. The actions planned for 2023 have been completed; more data will be transmitted in 2024 and 2025, as field inventories are carried out.

Status : In progress

2 BALANCE 2023

	No	Action	Status
Theme 1:	1	Environmental Effects Monitoring (EEM) - Cycle 4 and monitoring of metal concentrations in fish flesh from the Puvirnituq River	100%
	2	Monitoring the population and mercury concentration in the flesh of fish in Lac du Bombardier	100%
	3	Continuation of the register of caribou sightings in the Méquillon sector and inclusion of worker sightings	100%
Theme 2	4	Monitoring the free movement of fish in PNNi culverts	100%
	5	Monitoring Arctic char spawning grounds	100%
	6	Monitoring loggerhead damselfly colonies	100%
	7	Drawing up a wildlife protection plan and updating the Wildlife Management procedure	100%
	8	Installation of visual markers around plants with precarious status	100%
	9	Acquisition of skips that are less attractive to wildlife	100%
	10	Employee awareness workshops and capsules	100%
Theme 3	11	Progress of the Inuit Community Environmental Improvement Program (ICIEP)	In progress
	12	Installation of bird-scaring devices on MCPs	100%
	13	Creation of a weir downstream of the TR3 crossing	100%
	14	Reconstruction of the streambed and creation of a channel for fish passage at crossing TR2	100%
Theme 4	15	Information program in Inuit communities	100%
	16	Consultations on Phase 2 mining projects in PNNi - Wildlife component	In progress
Theme 5	17	Contribution to Caribou Ungava	In progress
	18	Support for MELCCFP representatives in locating a peregrine falcon	100%
	19	Sharing data with Projet Biodiversité Québec	Completed

3. OBJECTIVES 2024

The Plant and Wildlife Protection Plan (PPFF) is the wildlife management plan for PNNi. Thus, the next actions planned for 2024 will be in line with the objectives of the PPFF and the Nunavik Nickel Agreement. Preliminarily, the actions planned for 2024 are :

Theme 1: Improving knowledge assessment

- Maintenance of the caribou observation log and addition of the collection of observations from helicopter pilots present at the site.
- Completion of wildlife event reports for a wider range of observations, and putting these reports online to facilitate participation by CRI employees.

Theme 2: Preventing and protecting biodiversity

- Monitoring the free movement of fish in PNNi culverts
- Spawning ground monitoring
- Broadening the means of dissemination and the type of awareness-raising capsules (posters, television, etc.).
- Distribution of the updated Wildlife Management procedure
- Resumption of annual tundra scrap clean-up activity

Topic 3: Impact mitigation and compensation measures

- Continued development of the Inuit Community Environmental Improvement Program (ICCEP)
- Installation of additional bird-scaring devices at the Puimajuq and Ivakkak water-collection basins and reinstallation of bird-scaring devices.
- Biodiversity taken into account to a greater extent in the development of restoration plans

Theme 4: Stakeholder consultation, information and involvement

- Ongoing information program in Inuit communities
- More intensive deployment of Inuit community surveys with biodiversity components
- Invite one or more Inuit representatives to environmental monitoring activities
- Continued consultations on mining projects, including wildlife components

Theme 5: Support for research and development

- Continued partnership with Caribou Ungava
- Continuation of PAECI
- Additional data sharing with Projet Biodiversité Québec

APPENDIX: TSM STRATEGIC *FRAMEWORK Mining and the biodiversity conservation*

Towards sustainable mining development

Strategic framework

Mining and biodiversity conservation

Biodiversity ^{conservation}¹ is a commitment set out in the Guiding Principles of MAC's Towards Sustainable Mining initiative. MAC members recognize that a company's access to land and social license are intimately linked to its socially, environmentally and economically responsible practices, and that the analysis makes a compelling case for the industry to support biodiversity conservation. MAC members believe that mining, conducted in consultation with communities of interest, can coexist with biodiversity conservation. MAC members also believe that corporate adherence to the principles of biodiversity conservation is essential, and as such have agreed to the following commitments:

1. MAC members will make a positive contribution to biodiversity conservation at every stage in the life cycle of their mines.
2. MAC members will work with key communities of interest² to develop and implement responsible policies and practices to:
 - a. integrate the importance of biodiversity conservation, including respect for critical habitats, into mining activities, land-use planning and management strategies, and consider the option of not proceeding with certain mining projects;
 - b. assess and monitor the state of biodiversity throughout the life cycle of each mining project;
 - c. avoid, minimize, mitigate and/or compensate for any adverse effects on biodiversity;
 - d. improve, through research, information sharing and/or partnerships, the industry's understanding of and contribution to biodiversity conservation, science and traditional knowledge;
 - e. establish, fund and implement comprehensive reclamation plans that, as far as possible, restore the viability and diversity of ecosystems that will meet the needs of potential post-closure uses, while recognizing that mining activity can permanently alter the landscape, and that options for post-closure uses can be explored.

¹ The United Nations Convention on Biological Diversity defines biodiversity as "the variability among living organisms from all sources including, *inter alia*, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems". ² For example, governments, indigenous governments and communities, other local communities and conservation organizations.

may be taken into account in establishing restoration plans, as justified by the particular circumstances of each site.

3. MAC members are committed to transparency and public accountability on mining and biodiversity conservation issues.
4. Recognizing that protected areas contribute to biodiversity conservation, MAC members will respect the requirements for regulated protected areas and are committed to working with local communities of interest to develop transparent, inclusive, informed and equitable decision-making processes for the establishment of protected areas.
5. MAC member companies undertake not to conduct any exploration or extraction activities within World Heritage sites. All possible measures will be taken to ensure that pre-existing operations within World Heritage sites, as well as existing and potential operations in the vicinity of World Heritage sites, comply and co-exist with biodiversity conservation objectives.
6. MAC and its members will show leadership by disseminating this Strategic Framework on Mining and Biodiversity Conservation, and by encouraging other organizations to sign up too.

June 2007